



Evolutionary Computation

Assignment 8: Global convexity

Tests

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1 Introduction

The fitness-distance correlation (FDC) is a technique used to understand how good solutions relate to each other structurally. If solutions with similar structure tend to have similar quality, the search landscape is considered "easy"—local search (LS) can move smoothly from one good solution to another. If there is little relationship between structure and quality, the landscape is "difficult" or "rugged."

In this project, we study the landscape of the traveling salesman problem (TSP) by examining 1000 local optima found using simple LS. For each solution, we measure how similar it is to two reference points: (1) the best solution found, and (2) a very good solution produced by our best algorithm (ILS). We also compute how similar each solution is, on average, to all the others to understand the overall organization of the solution space.

We use two ways to measure similarity: counting shared edges (comparing tour structure) and counting shared nodes (comparing which cities are selected). We perform this analysis on two TSP instances using three different similarity metrics and two similarity measures, giving us 12 different comparisons that help us understand the problem's difficulty and structure.

You can see the repository of our assignment on github:

https://github.com/tbengric/EC_laboratory

2 Pseudocode

In this section, we will present the pseudocodes of similarity algorithms use to calculate the similarities. For this assignment, we will code in C++.

Common Edges

```
FUNCTION commonEdges(path1, path2):  
  edges1  $\leftarrow$  EMPTY_SET  
  edges2  $\leftarrow$  EMPTY_SET  
  
  FOR i FROM 0 TO LENGTH(path1)-2 DO :  
    a  $\leftarrow$  path1[i]  
    b  $\leftarrow$  path1[i+1]  
    IF a > b THEN : SWAP(a, b)  
    INSERT (a, b) INTO edges1  
  
  FOR i FROM 0 TO LENGTH(path2)-2 DO :  
    a  $\leftarrow$  path2[i]  
    b  $\leftarrow$  path2[i+1]  
    IF a > b THEN : SWAP(a, b)  
    INSERT (a, b) INTO edges2  
  
  common  $\leftarrow$  0  
  FOR edge IN edges1 DO :  
    IF edge IN edges2 THEN :  
      common  $\leftarrow$  common + 1  
  
  RETURN common
```

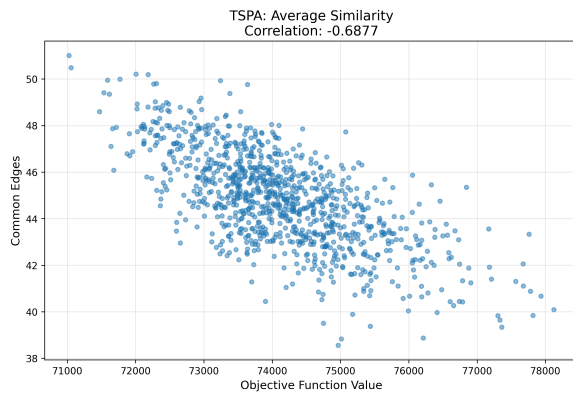
Common Nodes

```
FUNCTION commonNodes(path1, path2):  
  nodes1  $\leftarrow$  SET(path1)  
  nodes2  $\leftarrow$  SET(path2)  
  
  common  $\leftarrow$  0  
  FOR node IN nodes1 DO :  
    IF node IN nodes2 THEN :  
      common  $\leftarrow$  common + 1  
  
  RETURN common
```

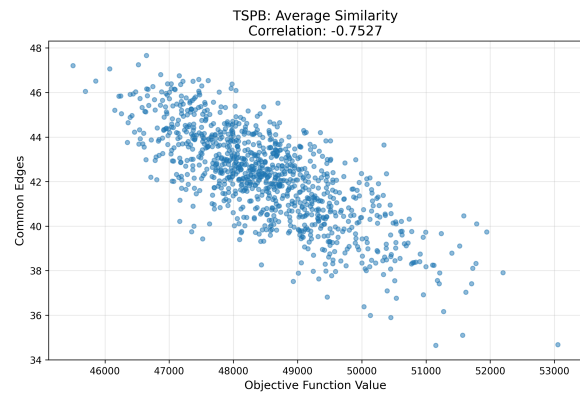
Correlation Coefficient

```
FUNCTION correlationCoefficient(x, y):  
  IF LENGTH(x)  $\neq$  LENGTH(y) OR LENGTH(x) = 0 THEN : RETURN 0.0  
  
  meanX  $\leftarrow$  SUM(x) / LENGTH(x)  
  meanY  $\leftarrow$  SUM(y) / LENGTH(y)  
  
  numerator  $\leftarrow$  0.0  
  denomX  $\leftarrow$  0.0  
  denomY  $\leftarrow$  0.0  
  
  FOR i FROM 0 TO LENGTH(x)-1 DO :  
    dx  $\leftarrow$  x[i] - meanX  
    dy  $\leftarrow$  y[i] - meanY  
    numerator  $\leftarrow$  numerator + dx  $\times$  dy  
    denomX  $\leftarrow$  denomX + dx  $\times$  dx  
    denomY  $\leftarrow$  denomY + dy  $\times$  dy  
  
  IF denomX = 0.0 OR denomY = 0.0 THEN : RETURN 0.0  
  
  RETURN numerator / SQRT(denomX  $\times$  denomY)
```

3 Plots

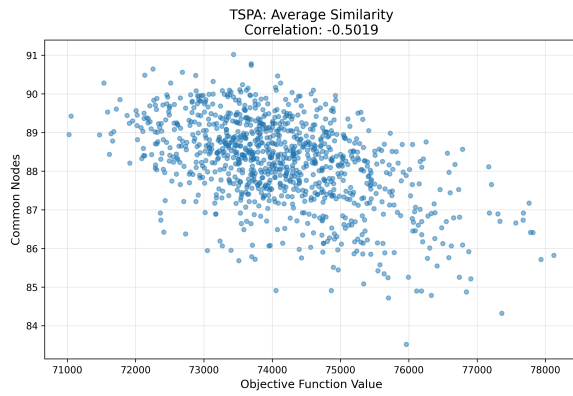


(a) TSPA

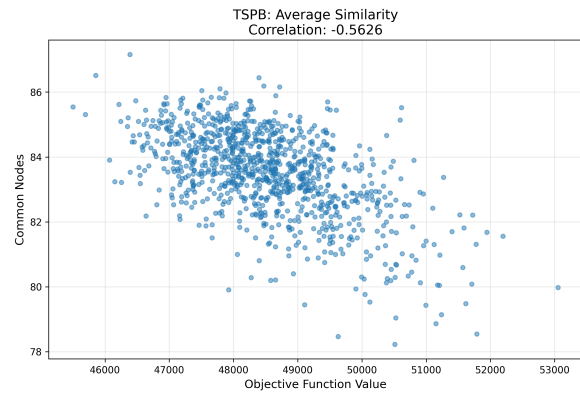


(b) TSPB

Figure 1: Average similarity Edges



(a) TSPA



(b) TSPB

Figure 2: Average similarity Nodes

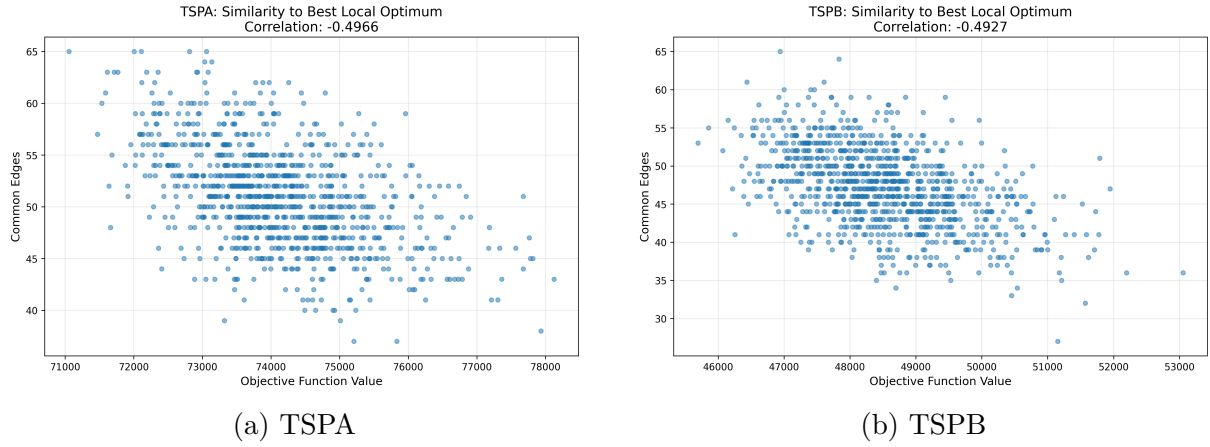


Figure 3: Best similarity Edges

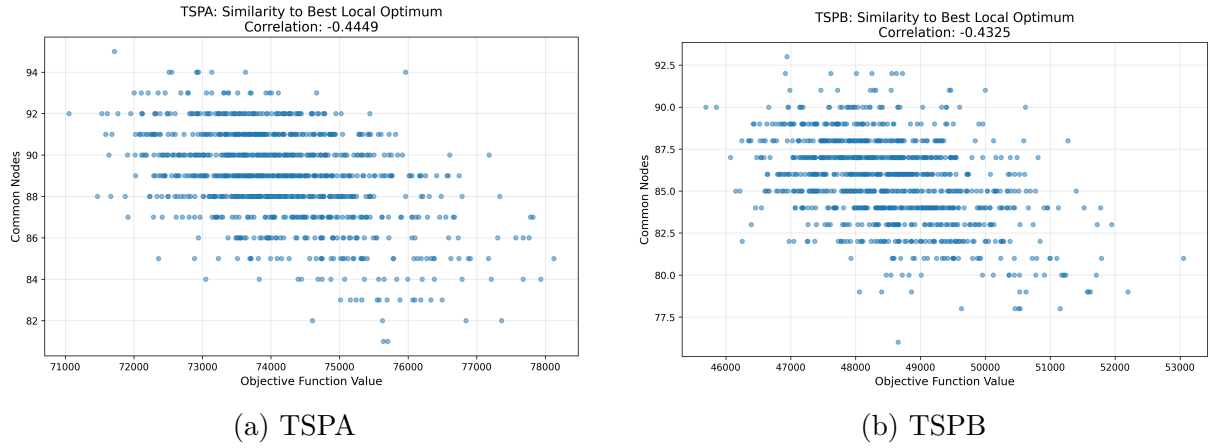


Figure 4: Best similarity Nodes

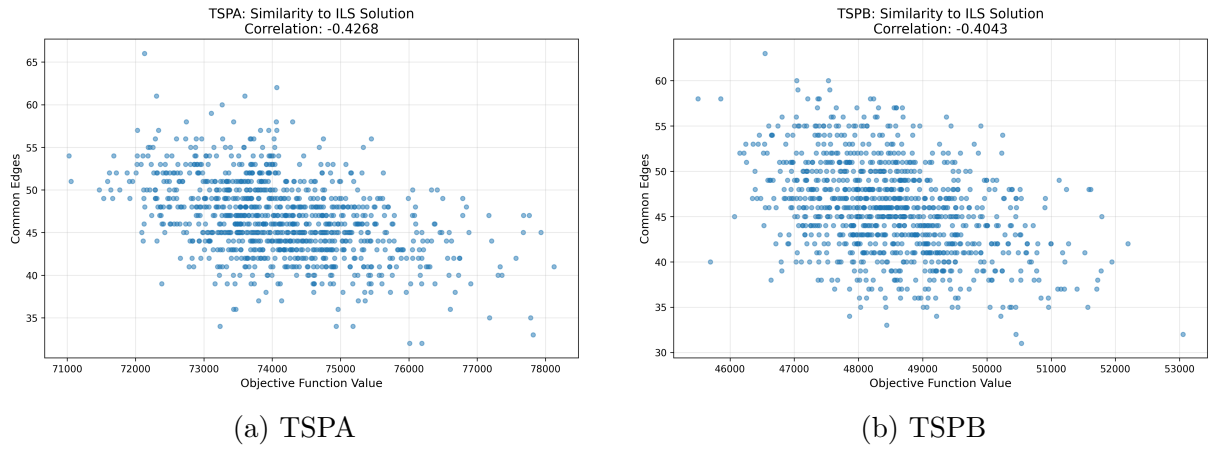


Figure 5: ILS similarity Edges

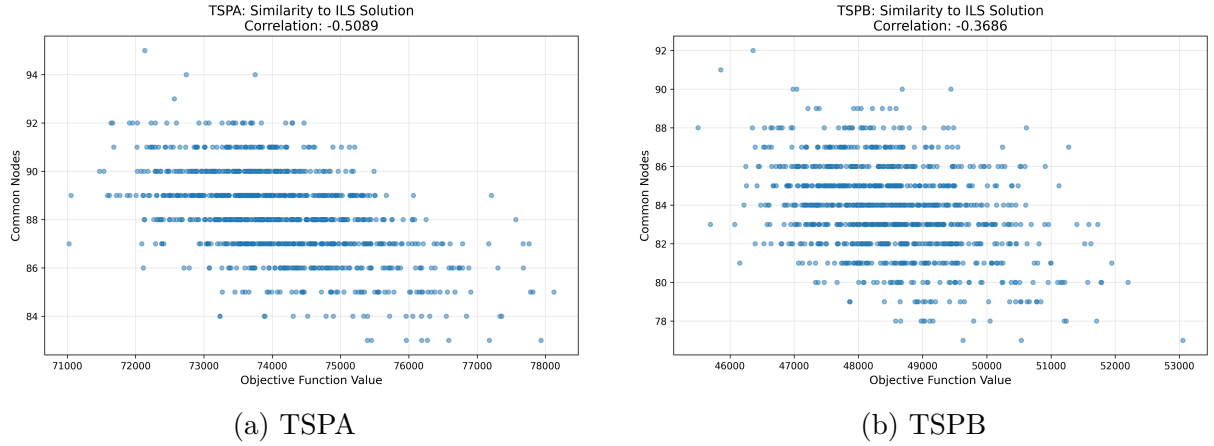


Figure 6: ILS similarity Nodes

4 Results

Similarity Type	TSPA	TSPB
Avg Similarity - Edges	-0.6877	-0.7527
Avg Similarity - Nodes	-0.5019	-0.5626
Best Similarity - Edges	-0.4966	-0.4927
Best Similarity - Nodes	-0.4449	-0.4325
ILS Similarity - Edges	-0.4268	-0.4043
ILS Similarity - Nodes	-0.5089	-0.3686

Table 1: Fitness-Distance Correlation coefficients for TSPA and TSPB

5 Conclusion

The analysis reveals strong negative correlations between solution quality and structural similarity, confirming a convex fitness landscape. Better solutions consistently share more edges and nodes with each other, making LS-based algorithms effective for these TSP instances. Edge-based similarity outperforms node-based similarity, suggesting tour connectivity is more important for characterizing solution quality.